

curriculum vitae of
Aakash Chowdhury

· RESEARCHER IN TECHNICAL SCIENCES INFORMATICS ·

RESEARCH INTERESTS: SIDE-CHANNEL ANALYSIS; CRYPTOGRAPHIC ENGINEERING;
STATISTICAL LEARNING; EXPLAINABLE AI FOR SECURITY EVALUATION

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EDUCATION

- Nov. 2020 – present **Ph.D.** in Technical Sciences Informatics UNIVERSITY OF KLAGENFURT, AUSTRIA
Doctoral research under Prof. Elisabeth Oswald on **statistical and machine-learning methods for side-channel security evaluation**. Research focuses on multivariate leakage certification, leakage detection, and profiled/non-profiled differential side-channel analysis of embedded devices.
Courses: Introduction to cybersecurity, Machine learning and deep learning, and Nonparametric Bayesian statistics.
Status: Expecting defence by Jan. 2027.
- Jul. 2015 – Jun. 2017 **M.Sc.** in Statistics (Grade: CGPA 5.132/6) UNIVERSITY OF CALCUTTA, INDIA
Important Courses : Predictive statistical modelling, Time-series analysis, Supervised and unsupervised statistical learning modelling, etc., with a specialisation in Advanced statistical inference.
Thesis: Simulating and modelling football results to forecast league standings in the English Premier League.
- April. 2012 – May. 2015 **B.Sc.** with Statistics Major (Grade: CGPA 8.8/10) PRESIDENCY UNIVERSITY, KOLKATA
Courses : Statistics, Mathematics, Economics, & Computer applications

PUBLICATIONS

IN CRYPTOLOGY:

1. **Aakash Chowdhury** and Elisabeth Oswald. **Multivariate Leakage Detection**. In *IACR Transactions on Cryptographic Hardware and Embedded Systems*, 2026, Volume 2. DOI: [10.46586/tches.v2026.i2.296-324](#).
2. **Aakash Chowdhury**, Carlo Brunetta, Arnab Roy, and Elisabeth Oswald. **Leakage Certification Made Simple**. In *Crypto 2024, organised by the International Association for Cryptologic Research (IACR)*. DOI: [10.1007/978-3-031-68391-6_13](#).

IN APPLIED STATISTICS:

3. Mahitha Karee, Sirisha R Gundabattula, Latha Sashi, Harikishan Boorugu, **Aakash Chowdhury**. **Prevalence of metabolic syndrome in women with polycystic ovary syndrome and the factors associated: a cross-sectional study at a tertiary care centre in Hyderabad, south-eastern India**. *Diabetes & Metabolic Syndrome: Clinical Research & Reviews*. Vol 14, Issue 4, 2020. DOI: [10.1016/j.dsx.2020.05.006](#).
4. Aishwarya Kanneganti, **Aakash Chowdhury**, Asma Sajid, and Latha Sashi. **Incidence and Risk Factors of Excess Gestational Weight Gain in Indian Women**. *The Indian Journal of Nutrition and Dietetics*. Vol 56, Issue 3, Sep 2019. DOI: [10.21048/ijnd.2019.56.3.23207](#).
5. **Aakash Chowdhury**. Best poster award on **Modelling and simulating football result for prediction of a soccer tournament**. *10th International Triennial Calcutta Symposium. Conducted by the University of Calcutta and Calcutta Statistical Association*. 27-30 December, 2018. DOI: [10.13140/RG.2.2.31794.41926](#).

ADVANCED TRAINING

- [enCRYPTON](#) summer school program on **Cryptographic Solutions for Privacy-Preserving Technologies** held at Sabanci University, Istanbul, Turkey, in September 2023.
- [CISPA](#) summer school 2022 on **Trustworthy Artificial Intelligence** held on September 5-9 at Saarbrücken, Germany.

PROFESSIONAL EXPERIENCE

Oct. 2020 – Present	University of Klagenfurt, Austria	PROJECT RESEARCHER
	Selected Responsibilities:	
	• Developed statistical and machine-learning methods for profiled and non-profiled side-channel security evaluation. • Contributed to Austrian and EU-funded projects: RampUp Cybersecurity, SEAL, SPyCODE (Quantitative and Probabilistic Security Analysis), enCRYPTON, and Explainable Leakage Assessment (EELA)	
Mar. 2023 – Sep. 2026	University of Klagenfurt, Austria	TEACHING ASSISTANT
	Selected Responsibilities: For summer semesters 23,24,26 and winter semester 25, I supervised students in Introduction to Cybersecurity , and Advanced Topics in Cybersecurity 1 & 2 .	
	• Teaching student in <i>Basic cryptography</i> and <i>System security</i>, supervised in SCA-related projects.	
Jul. 2020 – Sep. 2020	CAREADD, St. John's Research Institute, Bangalore (India)	SRF BIostatistics
	Selected Responsibilities: Support statistical analysis of genetic and imaging data related to diagnosis of ASD patients	
Nov. 2019 – Mar. 2020	Machine Intelligence Unit, Indian Statistical Institute, Kolkata	PROJECT LINKED PERSON
	Overview: Worked on a project named Development of GPGPU-based parallel algorithms for land cover classification of remotely sensing data through machine learning methods under Prof. B. Uma Shankar.	
Jul. 2017 – Aug. 2019	Fernandez Hospital Foundation, Hyderabad (India)	STATISTICIAN
	Selected Responsibilities: Contributes statistical expertise for the preparation of protocols for the DIPLOMATE OF NATIONAL BOARD and fellowship theses. Assist clinicians with hypothesis generation, sample size calculations, statistical analysis, and content writing in the scientific protocols	

TECHNICAL SKILLS

Programming Languages: Python, R (CRAN), C/C++, GNU Octave.

Scientific Computing in Python: NumPy, SciPy, Pandas, Numba, Scikit-learn, TensorFlow, [dcor](#).

Platform: Linux (Ubuntu, Debian, Xubuntu), Windows (10, 11)

Research & Development Tools: Git, Anaconda, Jupyter Notebook, RStudio, Vim, tmux.

Scientific Writing: LaTeX (TeX Live, Beamer, BibTeX)

Projects: [Leakage Detection Methods](#), [Leakage Certification](#), [uELMO](#), [EPL predictions](#), [COVID-19](#).

LANGUAGES & EXTRACURRICULAR ACTIVITIES

- Familiar with 4 languages: *Bengali* (Native), *English*, *Hindi*, and *Deutsch* (A2 level)
- Apart from research, I am also interested in indoor and outdoor activities, like cooking Indian cuisine or playing futsal, volleyball, and video gaming. Football and Cricket are two of my favorite sports.